

Standard Operating Procedure & Troubleshooting Manual

Equipment: Jacketed Semi-Batch Reactor R-101 (10 m³)

Document Ref: SOP-ENG-R101-REV6

Plant Location: Emulsion Plant (PVAc wood glue)

Applicability: Process Operators, Maintenance Technicians, Shift Supervisors

1. Equipment Description

Reactor **R-101** makes polyvinyl acetate (PVAc) emulsion for wood adhesive. Vinyl acetate monomer is fed gradually into hot water containing surfactant and initiator, so that it reacts as it is added. The polymerisation is **strongly exothermic** (about 90 kJ per mol of monomer). Heat is removed by cooling water in the jacket and by a reflux condenser. If monomer is fed faster than it reacts, unreacted monomer **accumulates** and can later react all at once, causing a temperature runaway.

2. Operating Envelope & Alarm Limits

Table 1: Operating envelope and alarm limits

Parameter	Normal Range	Warning	Critical Alarm	Trip / Action
Batch temperature	78 – 82 °C	above 84 °C	above 87 °C	90 °C: feed trip + short-stop injection
Cooling water inlet temperature	25 – 30 °C	above 32 °C	above 35 °C	Stop monomer feed
Cooling water flow	40 – 50 m ³ /h	below 35 m ³ /h	below 30 m ³ /h	Stop monomer feed
Monomer feed rate	350 – 450 kg/h	above 470 kg/h	above 500 kg/h	Feed pump trip
Agitator speed	80 – 90 rpm	below 75 rpm	below 60 rpm	Stop monomer feed
Free monomer (sample)	below 5 %	5 %	8 %	Stop monomer feed

Free monomer is sampled every 30 minutes during the feed. The short-stop is an inhibitor solution that stops the reaction.

3. Fault Matrix

Table 2: Faults, signs, causes and actions

Fault	Typical Signs	Likely Causes	Operator Action
FAULT-A: Cooling failure	- Batch temperature rising - Cooling water flow low or inlet temperature high - Agitator NORMAL	Cooling pump trip, cooling tower problem, control valve failure.	- STOP monomer feed. - Start standby cooling pump; check valve. - If batch is above 87 °C, inject short-stop.
FAULT-B: Monomer accumulation	- Batch temperature falls below setpoint while feed continues - Free monomer above 5 % - Followed later by a sudden temperature rise	Initiator feed stopped or weak, inhibitor in the monomer lot, feed rate too high.	- STOP monomer feed. - Do NOT heat the batch to 'catch up'. - Check initiator feed. - Restart only when free monomer is below 3 % and the engineer approves.
FAULT-C: Agitator failure	- Agitator speed or power drops sharply - Temperature probes disagree	Motor trip, coupling or shaft failure.	- STOP monomer feed. - Do NOT restart the agitator without the engineer.

	• Poor heat removal		3. Apply maximum cooling.
FAULT-D: Jacket fouling	• Batch-to-jacket temperature difference slowly increasing over many batches • Cooling valve near 100 % open	Scale in the jacket, polymer build-up on the reactor wall.	1. Reduce feed rate to keep temperature in range. 2. Schedule wall cleaning and jacket descaling.

4. Operating Rules

- On any high batch temperature, **stop the monomer feed first**.
- Never increase the feed rate to make up for lost time.
- Never restart a stopped agitator during a batch without engineer approval: sudden mixing of accumulated monomer can cause a runaway.
- Check the inhibitor certificate of each new monomer lot before use.
- Know where the short-stop injection is and wear full PPE when sampling.